



Is (Islamic) Occult Science Science?

Matthew Melvin-Koushki

ABSTRACT

More than any other object of historical and anthropological study, Islamicate occult science cuts to the quick of what it means to be modern, to be Western, to be scientific. Yet nowhere else are nineteenth-century colonialist metaphysics and materialist cosmology more firmly entrenched. The piecemeal, truncated study of “Magic in Islam” to date has thus often been pursued in service of either *scientistic* or *religionist* agendas, whereby magic can only ever be failed science or apolitical religion, and Islam can never be the West; Islamic Magic as simply Western and often imperial Science-and-Religion is thereby utterly disappeared from historiographical purview. This manifesto therefore proposes a way out of this dire epistemological and ethical bind. To restore Islamic Magic to its rightful place in Western intellectual and cultural history, especially history of science, we must take far more seriously the *panpsychist* cosmology on which it is predicated, and realize that our own reflexive materialism commits us willynilly to a colonialist agenda that is, ironically, both antireligious and antiscientific.

KEYWORDS

Occultism; colonialism; eurocentrism; materialism; panpsychism; scientism; religionism; experimentalism; science and empire studies; *mathesis*; psychophysics

The historical and anthropological study of Islamicate occult science,¹ like few other categories, plunges the modern researcher into a quandary equally epistemological and ethical. *Epistemological*, because it grossly violates the materialist-scientistic dogma still regnant among historians of science generally and Islamicists particularly—this at a time when some working scientists, philosophers of science, philosophers of mind and even theologians are defecting from materialism to panpsychism or dual-aspect monism as a more scientifically plausible cosmology.² (The latter cosmology, of course, from Pythagoras to the present, has historically driven Western occult-scientific theory and practice.³) And *ethical*, because it is a site where colonialist-orientalist dogma is still firmly entrenched. With few exceptions, premodern Muslim occult scieners—in extreme contrast to their Christian peers—figure explicitly or implicitly in recent and current historiography as superstitious, subrational natives fit only to be patronized, quarantined and ideally excised altogether.

This is not to suggest that anti-occultist historians of Islamicate science consciously pursue an antirationalist or morally heinous project; rather, the subfield as a whole has been heinously deformed and stunted from without, a casualty of the science and religion wars that shaped Euro-American modernity. Thus crippled by colonialist metaphysics, Islamicate science has all too easily been banished from the broader history of Western

This article has been republished with minor changes. These changes do not impact the academic content of the article.

science, which it may haunt only as outside influence; the polemical fiction first spun by Renaissance humanist anti-Arabists is now embraced as fact.⁴ After three centuries of physical and spiritual carnage, inaugurated by Reformationist witchhunting on the one hand and the Enlightenment boom in Cartesian mechanistic dualism on the other, *scientism* finally coalesced in the nineteenth century as an army mobilized against spirit, with *religionism* emerging in the twentieth as bulwark and citadel against that army. Neither side triumphing in their Wagnerian struggle, a *détente* was reached: henceforth the world would be divided into two kingdoms—whence Stephen Jay Gould’s doctrine of “non-overlapping magisteria.”⁵ (While this doctrine has been rejected as painfully, absurdly oversimplistic and patronizing by many Europeanists and working scientists and theologians alike,⁶ as well as increasingly by Islamicists,⁷ it still enjoys the full force of orthodoxy—if only implicitly—in the study of most things Islamicate.) With this seismic geopolitico-epistemological shift, this Endarkening erection of a great border wall between body and spirit, the domain of occult science was singled out for violent disincorporation, and its historically venerable and mainstream denizens made refugees, marooned or exiled to outer darkness.

A base definition is here in order: In the premodern Greco-Arabo-Perso-Latin scholarly tradition, those sciences routinely designated *occult* (Ar. and Per. *khafī*, Gr. *apókruphos*, Lat. *occultus*) were theorized and practiced as precisely that subset of the Hellenic (and Egyptian, Persian and Indic) natural and mathematical sciences designed to marry spirits to bodies and bodies to spirits, which is to say, to extrapolate from visible data (*ẓāhir*) to nonvisible (*bāṭin*), from matter to mind, human or otherwise.⁸ As such, even medicine (*‘ilm al-ṭibb*) could be and sometimes was classified among the occult sciences, in contrast to surgery (*‘ilm al-tashrīḥ*).⁹ To update this premodern formulation: they are the sciences concerned with *psychophysics*.¹⁰

Thus far epistemologically; sociologically, an equally common Arabic qualifier was *gharīb*, sciences the mastery of which is “unusual” and “rare,” which is to say, difficult, in that they presuppose competence in the natural and mathematical sciences more generally—as well as physical and spiritual purity. Thus roughly two to 15 percent of the most prestigious ulama captured in Arabic and Persian biographical dictionaries of the early modern era are identified as specialists in one or more of the occult sciences; a similar percentage are identified as physicians or hadith specialists.¹¹ By both definitions, in fact, epistemological and sociological, modern astrophysics or psychology too are often technically occult. That we now use this adjective exclusively as a slur on a par with “spooky,” and prefer instead neologistic prefixes like “dark,” “sub-,” “super-” and “para-” (as in dark matter, the subconscious, the supernatural, parapsychology) is of little epistemological consequence—beyond testifying to our continued belief in the magical power of naming!

Predicated on mind-matter monism in classic panpsychist fashion, occult science as an emic and an etic category is hence to be historically understood as simply *applied Neoplatonic-Aristotelian philosophy*, indeed a primary mode of Helleno-Islamo-Judeo-Christiane philosophical-scientific *praxis*. As such, it was embraced by scholarly and ruling elites throughout the premodern Western world—as yet unafflicted with Cartesianism—as the obvious, indispensable middle term between reason and revelation, mind and matter, the seen and the unseen, empire and eternity. Occult science disappears, in short, the strict modern dualist distinction between Science and Religion; such a powerful threat to the modern constitution must therefore be disappeared.¹²

And so it has almost totally been. As discrete academic fields, history of science and history of religion both—the very handles give them away—have been shaped from their inception by the Enlightenment- and especially Victorian-era vivisectionist conviction that the categories Science, Religion and Magic can be surgically separated and evolutionarily ranked, this in order to valorize the first, quarantine the second, damn the third, and vanish the sociopolitical context of all three.¹³ To be sure, in recent decades Europeanist historians of science and of religion, together with global historians, anthropologists and critical theorists, have made great strides toward exploding this colonialist engine of epistemic and ontological violence. Following Frances Yates (d. 1981), we may now speak of Western occult sciencers—including the great heroes of the Renaissance and the so-called Scientific Revolution (more properly a mathematical revolution), from Pico and Bruno to Bacon and Newton—as rational actors, may acknowledge the architects of scientific modernity as the committed occult sciencers they were.¹⁴ Indeed, modernity itself has been exposed as a construct of hypocritical sorcerousness, and disenchantment a powerful myth; magic is alive and well among colonized and colonizers alike.¹⁵ This upswell of revisionist and critical scholarship has culminated in the emergence of *Western esotericism studies* as a field in its own right, featuring dedicated academic units in Europe and scholarly organizations and journals on both sides of the Atlantic.¹⁶ And Euro-American history of science programs—although still overwhelmingly positivist and modernist in focus, and persistently eurocentric¹⁷—now sometimes include historians of occult science. Most significantly, the practice of modern science itself would seem to be nearing a historic inflection point: parapsychology (an unfortunate name) too now thrives as a more or less respectable if still marginal academic subfield, and has provided extensive experimental evidence for the reality, even normalcy, of such “unnatural” mind-matter phenomena as are perennially claimed by practicing occultists.¹⁸

But the *Islam* in Islamicate occultism remains a treacherous stumblingblock—or rather a deadfall. The qualifier immediately excludes the qualified from Westernness, from *rationality* itself, and vice versa: a disappearing act within a disappearing act. Islam as the *Occult West*; occultism as *oriental science*.¹⁹

Now Islamicists and Muslim scholars alike have long been ably combatting such extreme colonialist-orientalist perversions of intellectual and cultural history on many other fronts, from philosophy to poetry and astronomy to law; the wildly antihistorical thesis of “postclassical” Islamicate decline and decadence has finally been put to rest among specialists. (It remains virulent, needless to say, in contemporary popular and scholarly nonspecialist discourse.²⁰) But an exception continues to be made precisely for *occult science*, a first-order category of equal historical importance. This “occult exception” thus epitomizes the alchemically potent insidiousness of the colonialist pan-orientalizing project more broadly, the enduring power of the spell that is the *science-religion-magic* triad. While they often disagree sharply in all other respects, the overwhelming majority of contemporary Muslim scholars and scholars of Islam are united in their quiet elision—or strident disavowal—of this major component of Islamicate intellectual and cultural history. And their motives for doing so are at base the same, and indeed usually honorable: to defend Islamicate rationality against the declinists’ polemic, which triumphantly parades the ubiquity of occultism in Islam as proof of the latter’s essential unmodernity.²¹ Hobbled by apologetic, we have either sanitized beyond recognition or ignored altogether the Cusanuses and Keplers, the Brunos and Bacons of Islam.

Scientism, Religionism and the Birth of Modernity

To decolonize history of Islamicate philosophy-science, well-meaning Islamicists have thus been forced to choose between two equally pernicious approaches. In the first, magic—read: bad science and bad religion—is simply to be excluded from analysis to the extent possible; this is the *scientistic* strategy. In the second, magic remains bad science, but is magically made good religion by being disappeared into the catchall category of “mysticism”; this is the *religionist* strategy. To descientize or religionize occult science is therefore to defend, in high modernist and antimodernist fashion respectively, the honor of Islamic True Science or Islamic True Religion—and never the twain shall meet. In so subscribing to the colonialist triad, however, in so donning the dualist strait-jacket, historians of science and historians of religion alike have played into the polemicists’ hands—and too often been infected by polemic themselves.

That the situation is dire in history of Islamicate science is exemplified by the two most recently published synthetic overviews of the subfield. Egregiously, in his *Astronomy and Astrology in the Islamic World* (Edinburgh Univ. Press, 2016), Stephen Blake blithely dismisses the occult half of his subject as *pseudoscience* on the second page, proceeding to ignore it almost entirely for the rest of the book; and what little discussion there is is usually factually incorrect. Just as egregious, though otherwise far more balanced a survey, is Ahmad Dallal’s *Islam, Science, and the Challenge of History* (Yale Univ. Press, 2010), which refers four times in merest passing to astrology, explicitly of worth only as a foil for explaining the rise of astronomy, and mentions ubiquitous sciences like alchemy and magic but once—as being forever exiled to the shadowlands of Unscience by the scientific historian Ibn Khaldūn (d. 1406).²² This argument alone, formally identical to the long-outmoded claim that the great theologian Ghazālī (d. 1111) was responsible for the permanent crippling of philosophy within Islamdom, stands eloquent testament to the fact that the state of the field with respect to occultism could not be sorrier.²³ Needless to say, for modern historians to champion Ibn Khaldūn’s anti-occultist polemic simply because it seems to prefigure their own is sheerest anachronistic partisanship, is *pseudohistory*.

For recent studies have shown precisely the reverse to have happened: the notoriously unreliable Tunisian historian’s confused, fideist, rabidly antirationalist and ultimately abortive rant stands as index not of occultism’s demise, but rather its contemporary explosion in popularity among ruling and scholarly elites. Quite simply, the anti-occultists lost the debate, and lost it conclusively.²⁴ Ibn Khaldūn’s primary foes were a circle of Timurid and Ottoman occultists then wielding considerable influence at the Mamluk court, whom the Maliki judge attempted, unsuccessfully, to denounce as heretical sorcerers worthy of death. In setting him up as Hero of Science (bizarrely: this darling of French colonialism called for the outlawing of philosophy and various of the natural and mathematical sciences), Dallal is thus seduced into similarly denouncing lettrism (*‘ilm al-ḥurūf*)—Hebrew kabbalah’s coeval Arabic twin, and like it a science of outsize importance to history of Western science—as “cultic”!²⁵ This is not historiography, but heresiography.

Yet these surveys, it must be emphasized, accurately reflect the arguments and ethos of a majority of the field’s pillars to date. In his classic studies, Manfred Ullmann clearly assumes, for instance, that the natural sciences as practiced in the premodern Islamicate

world were so terminally infected by occultism as to render even “legitimate” sciences like medicine and astronomy crippled and finally abortive.²⁶ Fuat Sezgin, not content with mere scientism, attacks Ullmann for his sad acknowledgement of magic’s ubiquity in Islamdom, counterposing a thoroughlygoingly whiggish triumphalism that excises the religio-intellectual cancer that is occultism by acknowledging only the achievements of valiant Muslim thinkers laboring to preserve True Science—Greek, not eastern (Persian and Indian), and certainly not occult; thus only was Arabic science able to transmit the torch of the classical Greek heritage to Europe, subsiding into irrelevance after 1038.²⁷ Nor are Ullmann and Sezgin unrepresentative; even scholars like David Pingree, E. S. Kennedy, A. I. Sabra, George Saliba and Jamil Ragep, whose contributions to the history of Arabic science—including the history of astrology—are foundational and whose plumbing of the sources is more profound, often display the same tunnel vision that excludes certain bodies of data, especially occultist data, as irrelevant, even pesky.²⁸ While in most cases this defect does not impair the reliability of their findings in their narrow area of focus, which are frequently groundbreaking, it does result in tendentious, vertical narratives that float free of historical context,²⁹ as well as in whole volumes on the sciences in Islam that quietly censor the occult.³⁰

Shocking though the fact might be, it is hardly surprising, then, that more ink has been spilled to date on a single fourteenth-century anti-occultism polemicist, Ibn Khaldūn, than *all of the Islamicate occult sciences combined*.³¹ This colonialist historiographical artifact, in turn, continues to be reproduced and authorized even now by some Islamicists specializing in Magic in Islam; thus several chapters in the most recent edited volume on this theme, a massive German summa, reflexively present this would-be hunter of witches and burner of grimoires as the single greatest authority on occult science in Islamicate history.³² The sheer bizarreness of such fun-house distortions in the current historiography might best be appreciated by non-Islamicist American readers through the following analogy: this is like attempting to reconstruct a single, relatively minor battle of the Civil War (aka the War of Northern Aggression) exclusively on the basis of a small handful of Northern sources, then claiming this case study to be definitive of Southern History as a whole. Absurd, obviously. Even in that highly politically charged and sensitive field, no serious historian would dare to argue for the greater “objectivity” of hostile, outsider sources over defensive, insider ones, much less champion one side over the other—North against South, slaver against enslaved, European against African and Native. Yet Islamicists, it would seem, may do precisely that without professional repercussion, may freely take sides on the still raging battleground that is the historical and anthropological study of Magic.

Morally sound motives are thus not sufficient; methodological soundness is also required. That the science-historiographical mainstream so often strays beyond the bounds of scholarly rigor to engage in ahistorical polemic against its own subject is a methodological error surely disqualifying.³³

The religionist camp, arrayed staunchly against the evils of scientism, commit the equal and opposite error: polemicizing against Science, they recruit occult science—totally disappeared into “mysticism”—as a weapon useful in the assault on totalitarian modernity. Yet to be antimodern is to be modern, captive to a dialectical rictus; the ideological commitment here too has proved massively and terminally distorting. Members of the famed Eranos circle, the fathers of religionism include Gershom Scholem, Mircea Eliade and

especially Henry Corbin, each of whom seized upon rejected Religion as the existential core of humanness and mysticism as the core of religion in their respective areas of scholarly expertise.³⁴ Thus Scholem, who singlehandedly founded kabbalah studies (now a burgeoning academic industry in its own right), asserted the science to be the transcendent essence of Judaism; thus Corbin, who singlehandedly restored early modern Islamic philosophy (especially its renaissance in Safavid Iran) to the intellectual-historiographical map, asserted “Shi’i esotericism” to be the transcendent essence of Islam. Now the phenomenological rejection of history has its place in philosophical discourse, to be sure; but it makes a poor basis indeed for intellectual historiography, which rather charts the disjunctive evolution of conceptual archetypes amidst the rough and tumble of time and space. (Phenomenology tends to make for careless philology.³⁵) Similarly, a philosophical case can easily be made for *perennialism*; but the *history* of perennialism too is rampanly contradictory and divergent.

Lettrism is here again a case in point: Corbin and his disciples after him transmogrify this, the most Islamic of the occult sciences, encompassing various forms of letter magic and letter divination and rooted equally in the Quran and in Hellenistic precedent, into the epitome of perennialist Shi’i esotericism.³⁶ By the early modern period, it was indeed widely considered the perennial core of Alidism and Pythagoreanism in equal measure, a favored means—among Shi’is and Sunnis, scholars and sovereigns, sufis and scientists alike—of harnessing Imamic sacral power (*walāya*).³⁷ But in the fifteenth century it was reconceptualized as the primary Islamicate mode of *Neopythagoreanism*, and hence the Two Books doctrine, peculiarly Western, whereby the cosmos becomes a second scripture to be decoded, and magically recoded, by the self-theosizing scientist: *world as mathematical text*. As such, its burgeoning in Timurid Iran and Transoxania in particular was therefore the immediate intellectual and sociopolitical context, I contend, for the epochal mathematization of astronomy now so rightly celebrated by historians of science, and shown by them to be contiguous with, perhaps even an impetus for, the Copernican revolution in Europe.³⁸ The venerable *mathesis* narrative, which hails the early modern mathematization of the cosmos as the birthing grounds of scientific modernity itself,³⁹ is thus incoherent without reference to the Arabic astronomical revolution, and that revolution inexplicable without reference to the renaissance of lettrist Neopythagoreanism under the Timurid imperial aegis.⁴⁰

The Arabic science of letter-number, to no less degree than its Hebrew twin, would thus seem to be utterly indispensable to any history of *mathesis*; yet only kabbalah has been admitted to the narrative to date, and that only in its *Christian* reception.⁴¹ The guiding if implicit principle is clear: Jews and Muslims can only be unmodern mystics, for all that Jewish mysticism (never Islamic) may inspire modernizing Christian science. Put so baldly, such a formulation is patently egregious. But it captures fairly the dismal state of the field, divided and wartorn in the aftermath of the scientism-religionism contest. For their *détente* has proved methodologically disastrous: the phenomenological introversion and sacralization of occult science totally disappears it from physical and hence historiographical purview, which act of legerdemain has freed historians of science, manifestly relieved, to pursue a narrow, strictly materialist teleology neatly pruned of all such “cultishness,” to forget that *science is sacral*. (This truism, ironically, would seem to hold even more true of modernity than premodernity, when the scientific priesthood has largely displaced the religious.)

The mathematization of the cosmos so definitive of Western early modernity is thus inextricable from its sacralization, that is to say scripturalization, with occult science as middle term—a basic fact so far only recognized by historians of science with respect to Latinate Europe.⁴² Combined and weaponized, the occult exception and the European exception have served to utterly elide Islamicate occult science from the story of True Science. This potent, high-colonialist strategy not only makes impossible a history of the *practice* of Islamic philosophy;⁴³ it also renders suspect the very *history of modernity*.

An escape from this dialectical rictus was proposed by another eminent member of the Eranos circle: C. G. Jung. (It is often forgotten that this father of psychoanalysis was one of the most accomplished occultists of the twentieth century.) In particular, the Jungian doctrine of synchronicity as physical-metaphysical mechanism for divination—his focus was on the *I Ching* rather than its Western cognate, geomancy (*ilm al-raml*)—testifies to his efforts to synthesize quantum physics with psyche; like a good Neopythagoreanizing lettrist-kabbalist, he posited number as archetype marrying mind to matter. But Jung's approach to occult science was likewise interiorizing and ahistoricizing, and thus quickly commandeered by his disciples for religionist purposes.⁴⁴ Tellingly, psychoanalysis itself is now widely panned by psychologists as a pseudoscience.

Those historians who would be respectful of the premodern occult sciencers they study thus tend to fall into the trap of *psychologization*: they attempt to restore some measure of rationality to occultism by presenting it, *à la* Jung, as reflective of processes of mental-spiritual development.⁴⁵ But this, of course, merely reassigns complex, mainstream divinatory sciences like geomancy from history of science and history of technology to psychology and anthropology. The reflexive materialist assumption that mind-matter continuity is flatly impossible remains intact; magic can be at best a form of psychological warfare, never a means whereby mind can directly influence matter—the basic claim of all mages, past and present.⁴⁶ To psychoanalyze is thus to patronize, to know historical actors better than they can know themselves, to claim a superior level of psychohistorical development, to continue *la mission civilisatrice*.

Occult science, in short, represents an intractable historiographical problem equal parts philological and cosmological, emic and etic, epistemological and ethical. As such, its study is both professionally and personally hazardous to modern historians, who have therefore, understandably, closed ranks against Magic.

Philological Empiricism Saves the Postmodern Day

But there is always a third way. To deconstruct the modern ideological binary that has so colonized and perverted the historiography of science-religion, to escape the brutal matter vs. mind deadlock that is anti/modernity, to navigate the narrow channel between the Scylla of scientism and the Charybdis of religionism, we need only return to *radical philological empiricism*.⁴⁷ For the surviving Islamicate archive is massive, a scholarly bonanza of epic proportions, yet still, for ideological and cosmological reasons alone, almost totally untouched. Many hundreds of thousands of occult-scientific texts remain to be investigated. Let us therefore to the texts; and let us bracket out not the worldview of their authors, but our own, which will soon be supplanted in turn.

This is not to suggest that Islamicists have not been alive to the historical ubiquity of occult science in the surviving textual and artifactual sources from the very inception of

oriental studies as an academic field. As a rule, it seems, the more philological a historian's approach, the less likely they will arbitrarily ignore or actively suppress occult-scientific data. Philology alone does not entail occultophilia or prevent occultophobia, needless to say—hence the almost total lack of viable theoretical frameworks for the study of Islamic Magic, and the profusion of scientism and religionism in their stead, over the last century and a half. Nevertheless, a certain fascination among orientalists with matters occult is evident from the mid-nineteenth century onward, whose philological labors have provided a solid (if very partial and piecemeal) base for the study of the same going forward.⁴⁸ More recently, the empirical contributions of Emilie Savage-Smith, Charles Burnett, David Pingree, Richard Lemay, Manfred Ullmann, Toufic Fahd, Anne Regourd, Živa Vesel, Cornell Fleischer, Daniel Martin Varisco, Pierre Lory and Constant Hamès in particular have been formative.

But that base, like the study of Islam as a whole, remains largely arabocentric, classicocentric and/or “Middle East”-centric—artifacts all of colonialist metaphysics and its cult of origins.⁴⁹ It is only in the last decade that scholars (including historians of science like George Saliba, Jamil Ragep, Sonja Brentjes, İhsan Fazlıoğlu, Robert Morrison, Justin Stearns and Nahyan Fancy) have begun to systematically attempt to escape this plague of “centritis”; monographs on matters “postclassical”—which is to say, *early modern*—suddenly have a market.⁵⁰

Four recent or forthcoming edited volumes are here indicative of this seachange in the field, all of which reject antihistorical polemic and Whig teleology in favor of rigorous philology, and together geographically span from the Atlantic Ocean to the Pacific and temporally from the 7th century to the 21st: Matthew Melvin-Koushki and Noah Gardiner's *Islamicate Occultism: New Perspectives* (special double issue of *Arabica*, 64, nos. 3–4 [2017]); Nader El-Bizri and Eva Orthmann's *The Occult Sciences in Pre-Modern Islamic Cultures* (Beirut: Orient-Institut Beirut, 2018); Liana Saif, Francesca Leoni, Matthew Melvin-Koushki and Farouk Yahya's *Islamicate Occult Sciences in Theory and Practice* (Leiden: Brill, 2020); and Sonja Brentjes's *Routledge Handbook of Scientific Practices in Islamic Societies* (New York: Routledge, 2020). These volumes, quite simply, are the first in the field to systematically take seriously the occult nature of science and the scientific nature of the occult in a way that is now standard in Europeanist historiography.⁵¹ Equally notably, these volumes also reflect the current burgeoning of interest in occultism among junior Islamicist historians and anthropologists in particular.⁵² A fifth volume, Sebastian Günther and Dorothee Pielow's *Die Geheimnisse der oberen und der unteren Welt* of 2019, must also be added to this list; although, as noted, it generally theoretically falls prey to the scientism vs. religionism deadfall, its sheer size (almost 700 pages!) and unprecedented geographical, temporal and disciplinary scope likewise herald the emergence of Islamicate occultism studies as a major new front in Western intellectual and cultural history.

As the first of their kind, however, such studies must remain strictly preliminary: the histories of such seminal Islamicate natural-mathematical sciences as astrology, alchemy, lettrism and geomancy have *not yet been written*, and so can hardly be summarized here, and even the most seminal figures and texts remain almost or entirely unstudied.⁵³ Vast lacunae remain. A staggering amount of basic philological spadework therefore lies between us and even a general sense of the place of Islamicate occult science in history of science-and-religion; we are at least a century or two behind our Europeanist colleagues in this respect. We still await our Thorndike, to say nothing of our Yates.⁵⁴

The Occult Sciences in Classifications of the Sciences

Precisely how much work remains to be done? What proportion of the Islamicate natural and mathematical sciences were formally classed as occult?

My own impression, based on my preliminary survey of Arabo-Persian classifications of the sciences (sg. *taṣnīf al-ʿulūm*) produced from the tenth to the seventeenth century—the first in the field, astonishingly if symptomatically, to deal with post-Mongol Persianate developments—, is that the occult sciences constituted roughly *half* of the Islamicate natural and mathematical sciences for a full millennium, right down to the colonial rupture.⁵⁵ Naturally, this percentage varies widely from century to century, from region to region and from author to author. But even manifestly anti-occultist encyclopedias are obliged to include many occult sciences as a matter of course; and the most influential classifications of the sciences in the later Arabo-Persian tradition are, *without exception*, heavily pro-occultist.

While occult sciences were perennially popular at court from the Umayyads onward, moreover, the fourteenth century, riven open by the apocalypse that was the Black Death, stands as a major inflection point in the history of Islamicate science and empire in tandem. For it was then that occultism, together with sufism and Alidism, was made central to the new, experimentalist modes of millennial sovereignty and philosopher-kingship being developed in Mamluk Egypt and Ilkhanid Iran in particular, and fully realized over the next three centuries in the form of competing Timurid, Aqquyunlu, Safavid, Mughal and Ottoman imperialisms, each pressed into ever more radical avenues of ideological experimentation by their common Turko-Mongol Perso-Islamic rivalry. The post-Mongol, early modern era is, quite simply, the golden age of Islamicate occultism, and the occult-scientific tenor of early modern Islamicate empire without precedent and parallel in Western intellectual-imperial history.⁵⁶ (That the same era was also the golden age of Judeo-Christianate occultism is not an accident.) In the process, the *gharīb* in *al-ʿulūm al-gharība* thus became a sociological descriptor meaning not “strange” but *elite*.

Unsurprisingly, this seachange is heralded by the two most influential classifications of the sciences ever produced in the Islamicate encyclopedic tradition as a whole, Mamluk and Ilkhanid respectively:

- (1) *Guidance for the Seeker of the Sublimest of Goals* (*Irshād al-qāṣid ilā asnā l-maqāṣid*), in Arabic, by Ibn al-Akfānī (d. 1348), a Cairene physician-alchemist who perished in the plague; and
- (2) *Jewels of Sciences Delightful to Behold* (*Nafāyis al-funūn fī ʿarāyis al-ʿuyūn*), in Persian, by Shams al-Dīn Muḥammad Āmulī (d. 1352), professor (*mudarris*) at Sultaniyya under Öljeytū (r. 1304-16) and Abū Saʿīd (r. 1316-35).

Following a strict Avicennan schema,⁵⁷ the first classifies all occult sciences as derivative natural sciences, which include:

Medicine (*ʿilm al-tibb*), veterinary medicine and falconry (*ʿilm al-bayṭara wa-l-bayzara*), physiognomy (*ʿilm al-firāsa*), oneiromancy (*ʿilm taʿbīr al-ruʾyā*), judicial astrology (*ʿilm aḥkām al-nujūm*), magic (*ʿilm al-siḥr*), talismans (*ʿilm al-ṭilasmāt*), illusionism (*ʿilm al-simiyāʾ*), alchemy (*ʿilm al-kīmiyāʾ*), agriculture (*ʿilm al-filāḥa*).

As Ibn al-Akfānī clarifies, these natural sciences treat of either simple bodies or composite bodies or both. Thus astrology treats of celestial simple bodies, and talismans treat of terrestrial or elemental simple bodies. As for composite bodies, illusionism treats of those that have no constitution (*mizāj*), while those that have constitutions are further subdivided into those that do not have souls (alchemy) and those that do have souls; here there are either insensible (*ghayr mudrika*) souls (agriculture) or sensible (veterinary medicine for animals, medicine for humans). Physiognomy then inquires into the inner states of rational human souls while they are conscious, and oneiromancy does the same while they are unconscious. That science which treats of both simple and composite bodies is magic.⁵⁸ He posits, in short, an *astrology-talismans-magic continuum* as the very backbone of natural philosophy and psychology both, running the epistemological-ontological gamut from celestial simple bodies to terrestrial or elemental composite bodies, and allowing the competent philosopher-scientist experiential control of the cosmos. By this standard, even agriculture may be considered an occult science.

Where this Mamluk offering is highly succinct and dry, however, its Ilkhanid counterpart is expansive, gorgeous and epistemologically revolutionary. Therein Āmulī too employs an Avicennan schema, synthesizing it with the then-standard basic division between the Islamic or latter-day sciences (*‘ulūm-i avākhir*) and the philosophical or ancient sciences (*‘ulūm-i avā’il*), and covers an unprecedented number of both. Yet he also seeks to transcend both schemas to move toward a new, more universal conception of knowledge—one in which, crucially, occult sciences like geomancy are to be reclassified as *mathematical*. The various branches of the natural and mathematical sciences are as follows:

Branches of natural science (*furū‘-i ṭabī‘ī*): medicine (*‘ilm-i ṭibb*), alchemy (*‘ilm-i kīmiyā*), letter magic (*ṣīmiyā*), oneiromancy (*‘ilm-i ta’bīr*), physiognomy (*‘ilm-i firāsāt*), judicial astrology (*‘ilm-i aḥkām-i nujūm*), active properties (*‘ilm-i khavāṣṣ*), physiological professions (*‘ilm-i ḥiraf al-ṭabī‘a*) (these including the two divinatory techniques of scapulomancy (*aktāf*) and spasmatomancy (*ikhtilāj-i a’zā’*)), breath control (*‘ilm-i dam*), meditation and visualization (*‘ilm-i vahm*) (*dam* and *vahm* being Indian yogic sciences).

Branches of mathematical science (*furū‘-i riyāzī*): planetary theory (*‘ilm-i hay’at*); optics (*‘ilm-i manāẓir*); the Middle Books (*mutavassīṭāt*); calculation (*‘ilm-i ḥisāb*); algebra (*‘ilm-i jabr u muqābala*); surveying (*‘ilm-i masāḥa*); constellations (*‘ilm-i ṣuvar-i kavākib*); star tables, ephemerides and astrolabes and other observational instruments (*‘ilm-i arqām u a’māl-i zīj u taqvim u ašturlāb u ālāt-i raṣadī*); administrative geography (*‘ilm-i masālik u mamālik*); magic squares (*‘ilm-i vafq-i a’dād*); geomancy (*‘ilm-i raml*); mechanics and wondrous devices (*‘ilm-i ḥiyāl*); games of strategy (*‘ilm-i malā’ib*).

Here too natural science is almost entirely occult; but the *Jewels of Sciences* further testifies to the progressive “mathematicalization” of the occult sciences—an epochal development, peculiar to the Persianate east, that must be considered central to the *mathesis* narrative going forward. Equally epochally, it is also the first encyclopedia to register the contemporary *sanctification* of occult science, and lettrism in particular; Āmulī posits the science of sufism (*‘ilm-i taṣavvuf*) the supreme Islamic science, equal in importance to all the other religious and traditional sciences combined, and lettrism the supreme sufic science, that is, the science of *walāya*.⁵⁹

Building on this Mamluk-Ilkhanid precedent, the leading occult sciencers of the next century—Timurid and Ottoman scholars radiating from Mamluk Cairo, and

calling themselves, significantly, the New Brethren of Purity—synthesized the parallel processes of mathematicalization and sanctification to reconceptualize lettrism as the supreme *Alid-Pythagorean universal-imperial* science, a total physics-metaphysics allowing the scholarly or royal adept comprehensive access to and experiential control over the various levels of being.⁶⁰ Also in Mamluk Cairo, alchemy too was rendered a universal *mathematical-natural-religious* science by Aydemir al-Jildakī (fl. mid-14th c.), the most important and prolific Arabic alchemist in the Western tradition after (Ps.-)Jābir b. Ḥayyān—and, of course, almost totally unstudied.⁶¹ The mathematicalization-sanctification of occult science generally and lettrism and alchemy particularly continued apace in the persophone world over the next centuries, such that Safavid encyclopedias—even more heavily Neopythagorean-occultist in tenor—seize upon these two sciences as both the primary mode of philosophical praxis and the best means of harnessing *walāya*—which is to say, of *magically incarnating the Imams*. ‘Alī b. Abī Ṭālib, inventor of letter divination (*jafr*), is now hailed as chief imperial mathematician and occult technologist of Islam.⁶²

For some early modern encyclopedists, moreover, the sanctification of certain natural-mathematical occult sciences was so total as to justify their reclassification as religious sciences. That is to say: the occult sciences, concerned with mind and matter in equal measure, became the only disciplines to transcend the otherwise strict epistemic and generic divides between the rational (*‘aqlī*) and traditional (*naqlī*), or ancient Hellenic and modern Islamic, sciences.

A case in point is the Ottoman polymath Taşköprüzāde Aḥmed (d. 1561), who in his comprehensive *Key to Felicity and Lamp to Mastery* (*Miftāḥ al-sa‘āda wa-miṣbāḥ al-siyāda*)—a massive Arabic expansion on Ibn al-Akfānī and primary model for all subsequent Ottoman encyclopedias, including in the first place Ḥājji Khalifa’s (d. 1657) landmark *Removing Uncertainty as to the Titles of Books and Disciplines* (*Kashf al-ẓunūn ‘an asāmi l-kutub wa-l-funūn*)—structures, for the first time in the Arabo-Persian tradition, the totality of Islamic scientific and religious knowing according to an explicitly *lettrist* schema. At the same time, he classes lettrism not as a mathematical science but as a religious science, and specifically as a subset of quranic exegesis (*tafsīr*): the surest route to the heart of scripture is *magic*.⁶³ The Two Books doctrine here remains intact; the emphasis, however, is on the primacy of the Quran as model of reality, rather than on that model’s call to mathematize the cosmos. This shift of emphasis reflects the larger Ottoman scholarly project to *de-mathematize* astronomy and return occult sciences like astrology and geomancy to the natural sciences, apparently as a means of differentiating Ottoman learning from its Timurid-*cum*-Safavid model and rival, utterly committed to an equally sacralizing, magical *mathesis*. For Taşköprüzāde’s Safavid peers went to the opposite extreme: they singled out lettrism and alchemy in particular as the two ultimate mathematical and Hellenic-Islamic-Imamic, which is to say universal, sciences.⁶⁴ In both Persianate cases, however, the first Sunni and the second Shi‘i, the moral cannot be more explicit—nor more divergent from contemporary developments in Inquisition-Reformation Europe: *Islam is to be lived as magic*.

While research into the Islamicate encyclopedic tradition is in its infancy, in sum, the overall trend from the medieval to early modern periods is clear. The natural, the mathematical and even the religious sciences became increasingly disproportionately *occult*;

and the greatest metaphysicians of early modern Islamdom were often also its greatest mages.

But is this rather extreme disproportionality reflected in the surviving textual archive? It would seem so, though again, given the almost total elision of occult science from Islamic intellectual and cultural history to date, it is impossible to say at this stage. Not only do many collections (especially but not only private) remain uncatalogued or very poorly so, or for political reasons are virtually inaccessible; colonialist ideology, whether French, English or Soviet, which dismissed all occult sciences as but the superstition of the natives and collapsed them as a category into “mysticism” or “shamanism,” and Islam itself into scripturalism and legalism, resulted in the widespread suppression and institutionalized destruction of the same in colonized societies—which suppression continues with equal vigor even in the postcolonial, autocolonial present.⁶⁵ In Egypt and Iran in particular—precisely the epicenters of the occultist renaissance just described—, most of the wealth of occult-scientific manuscripts is currently strictly off-limits to researchers. And in the more accessible collections of western Europe and North America, such texts tend to be significantly underrepresented as well as routinely miscataloged, for reasons that will now be obvious (the collections at Princeton and the Wellcome Institute being notable exceptions).

Despite such major logistical challenges, or rather because of them, we may and must denounce the impression given by the state of the field, to wit, that astronomy and medicine alone are the historical center of Islamic True Science, and neither in any way occult, as being certainly wildly false. To maintain this fiction is to perjure an astonishingly rich archive, is to know the natives better than they can know themselves. A fairer history of science in Islamicate societies—one that at least aspires to rough proportionality to the extant textual and material record, severely distorted and decimated as it is—would by rights be at least *half occult*. That such a history remains inconceivable means, quite simply, that Islamicate intellectual and cultural history remains at least half unwritten.

Occult-Scientific Experimentalism as Early Modernity

The historical and anthropological study of Islamicate occult science is not simply an efficient means of exploding colonialist-orientalist chestnuts as to the eternal unmodernity of Islam, however; it is also a project that cuts to the quick of *what it means to be modern, to be Western, to be scientific*.

For our occult sciencers constantly testify to the *empirical* nature of their operations. That is to say, Islamicate early modernity, like its Christianate parallel, is characterized by an increasingly heavy emphasis on the experimentally verifiable nature of science, occult and otherwise. Here the central Arabic terms are *tahqīq*, “independent verification,” and *tajriba*, “experiential assaying”; the latter’s semantic evolution closely parallels that of its Latin cognate *experientia*.⁶⁶ This, it must be emphasized, represents a significant departure from Aristotelian precedent, wherein *empeiria* is a mode of general knowledge considerably inferior to *epistēmē*, “science.” The empirical nature of occult science was already asserted by such authorities as Abū Ma’shar Balkhī (d. 886), putative father of conjunction astrology, who influentially declared his discipline to be a science founded on analogical reasoning (*qiyās*) and experience (*tajriba*), the latter entailing observation (*mu’āyana*) and direct discovery (*al-wujūd al-ḥādīr*).⁶⁷ An identical argument is made for medicine

—sometimes classified as an occult science—by Ibn Hindū (d. 1019 or 1029), who rather uses the terms *raṣād* and *mushāhada* for observation and personal inspection respectively.⁶⁸ And Ibn Sīnā (Avicenna, d. 1037) himself, that second Aristotle, asserted *tajriba* to be superior to Aristotelian inductive reasoning.⁶⁹

By the early modern period, however, such categorizations had become standard for the occult sciences as an epistemological unit; geomancers, lettrists, astrologers, alchemists and physicians now insist without exception on the *mujarrab* nature—the *reproducibility*—of their findings and procedures, now lay exclusive claim to *taḥqīq*.⁷⁰ Equally importantly, the same period saw an explosion in the production of Arabic, Persian and Turkish occult-scientific manuals that explicitly invite the reader to master their contents independently of a teacher. This too represents a major departure from medieval precedent, whereby occult science was as a rule sociologically occulted through esotericist textual practices meant to exclude noninitiates.⁷¹

Crucially for history of science, such an occultist-experientialist ethos is highly reminiscent of Baconian experimentalism, that basis of scientific modernity—wherein natural-mathematical science was just as frequently occult. Indeed, Francis Bacon (d. 1626) himself explicitly pursued the rehabilitation of what he imagined to be ancient *Persian* magic, as well as alchemy, astrology and various forms of divination, as the core of his new Protestantizing form of natural philosophy: he invoked magic not as superstition but as superstition's cure.⁷² As in Latinate Europe, then, occult science was widely and explicitly embraced throughout the Islamicate and especially Persianate world as a means of demonstrating the validity of Neoplatonic-Aristotelian and especially Neopythagorean physics-metaphysics, and hence a primary driver of *mathesis*. Its investigation as a primary, and perhaps *the* primary, mode of philosophical-scientific experimentalism in post-Avicennan Islamicate intellectual and cultural history is therefore a desideratum particularly pressing.⁷³ The fact that occult science continues to be a site of experimentalism even in hypermodernized Islamicate societies today makes it an equally revealing category for anthropologists.⁷⁴

But unlike in Latinate Europe, it also became foundational to early modern Islamo-Mongol Alexandrian-universalist *imperialism*. Which is to say: Bacon's famous dictum—*knowledge is power*—would seem to have been taken more seriously, and to much greater political effect, by early modern *Muslim* ruling elites and the occult sciencers that served them; contemporary Christian imperial ideologies could hardly scientifically or materially compete.⁷⁵

For not only does occult *science* naturally generate, and indeed require, occult *technology*; in the history of Magic, pragmatic concerns usually long precede theoretical or philosophical ones. At the same time, it is the vigorous dialectic between grubby, experimentalist praxis and highflown theory that historically produced ever more sophisticated forms of occult technology, which, when effective, were held up in turn as scientific proof for the validity of panpsychist (Neoplatonic-Neopythagorean) cosmology. The *talisman* (*ṭīlasm*) is a case in point: an astral-magical device with deep roots in Hellenic-Egyptian antiquity, it remained fairly rudimentary until the tenth century, when the seminal *Goal of the Sage* (*Ghāyat al-ḥakīm*, famed as the *Picatrix* in Latin Christendom) provided the first comprehensive system of correspondences across multiple ontological categories as a rigorous basis for talisman production, thereby greatly extending its scientific and philosophical salience alike.⁷⁶ A second revolution occurred from the thirteenth century

onward, though only in the Islamicate context, when the Arabic talisman was reformulated in simultaneously *mathematical*, *religious* and *imperial* terms—a form of technology for achieving both human theosis and world empire.

Not unlike the practice of modern laboratory science, moreover, the premodern production of such occult-scientific technologies required as a rule the construction and maintenance of highly controlled and artificial settings. But in extreme contrast to their modern cognates, those premodern settings had at least as much to do with the ritual control of human—and nonhuman—modes of *consciousness* and *imagination* as with the manipulation of matter. Even in the alchemist's laboratory, featuring every manner of specialized and costly contraption, the elevation and purification of mind was normally prerequisite for the elevation and purification of matter.

Their dogmatic disavowal of mind aside, however, modern scientists are closely allied to their premodern occult-scientist forebears in this most crucial respect: both privilege the *extreme condition*; both are initiates into the *traumatic secret*.⁷⁷ Hence the trillions of dollars that are now spent on traumatizing “nature,” from atoms to human animals and everything in between (angels and planetary intelligences are curiously exempt), or on observing, with increasing clarity, the extreme and unthinkable ancient traumas of cosmic evolution.⁷⁸ Hence premodern Muslim occult scientists' stringent requirement that one observe total seclusion, ritual purity, constant vigil, frequent prayer, astrological obedience and a minimal, vegetarian diet when carrying out an operation, for weeks, months or even years on end—this to achieve an altered, extreme state of consciousness in which experimental designs, or even entirely new sciences, may be delivered through dreams and visions. And hence the perennial insistence by practicing mages of all traditions as to the particular magical effectiveness of operations that are grisly, including both animal and human sacrifice. (That the Islamo-Christian occultist renaissance from the late fourteenth century onward was a direct product of the plague-induced death of up to half of the human race across Afro-Eurasia—an extreme trauma, to say the least—is no accident.⁷⁹) In all cases, to do proper science the faithful scientist must be subject to *wonder* (*ta'ajjub*), in some cases traumatically so.

Such premodern-modern continuities aside, this new brand of Islamicate imperialist-occultist experimentalism, despite its historical success and vigor over vast swathes of the early modern Afro-Eurasian ecumene, and the robust parallel it makes to contemporary Christianate developments, *did not lead to us*. As such, it provides historians of science a crucial scientific control for the study of Western early modernities as a distinct if luxuriantly divergent cultural-genetic group;⁸⁰ this genealogical approach in turn will allow us to banish the specter of Whig history that continues to structurally haunt the field, still doggedly eurocentric and scientistic, the last decades of critical theory all notwithstanding. Shall we historians of science-and-religion not finally heed Darwin, Christian Hero of Science, who declared that *the tree of evolution forever branches, and has no point*?⁸¹

Conclusion

In recent decades it has become acceptable for Europeanist historians to patronizingly acknowledge that occult science was considered to be simply good science by benighted premoderns—and in some cases even accidentally productive of what we moderns consider to be science. Islamicists remain well behind even this modest curve. The recent

historiographical invention of the “Scientific Revolution”⁸² is a case in point: its greatest heroes (like their unstudied Muslim peers) were committed occult scieners to a man. But was Islamicate occult science Science too?

Yes, naturally—and mathematically. We cannot historically explain *mathesis*, a project unique to the West from Pythagoras to the present, nor the rise of early modern experimentalism, the very matrix of scientific modernity, without it. This question is therefore the most efficient means of decolonizing Islamicate intellectual and cultural history specifically and Western generally—as well as the modern practice of science itself, still predicated on the Science-Religion-Magic triad. That is to say, Magic offers us the most efficient methodological escape from orthodox positivist and religionist teleologies alike, a potent peepstone onto alternate Western (Hellenic-Islamic-Judeo-Christian) early modernities that would have led, perhaps, to more sustainable modernities. It remains the rejected Other upon whose undead frame our peculiarly narrow yet totalitarian Euro-American colonialist-capitalist brand of modernity has been and continues to be constructed. That Magic is often simultaneously Religion—which is to say, concerned as much with mind as with matter—as well as Empire—which is to say, concerned as much with power as with knowledge—emphatically does not exclude it from history of Science. Likewise, that Islam as imagined and lived was (and still is) often tantamount to Magic emphatically does not exclude either from Western Civ. As a host of early modern Muslim scholarly and ruling elites roundly declared: *Islam is magic; magic is science; science is empire*.

All well and good; the ethical imperative demands that we decolonize the past. And Magic, given how consistently it has been invoked by pro- and anticolonialist forces alike, is a particularly strategic historiographical and anthropological category to this end. But the question we still may not ask: *Is occult science Science?*

Let us historians of Western science, traumatized survivors of the Science-Religion wars that we are, therefore finally historicize ourselves. Let us break the binding spell of scientism and religionism alike. And let the Islamicists among us cease occulting the half of our astonishingly rich Arabo-Persian archive (to say nothing of other “post-classical” vernaculars like Turkish, Urdu, Malay or Swahili) that happens to be technically occult. Let us too be less patronizing, less colonialist, less materialist, less polemical, and more humble, more modern, more pragmatist, more empirical, if only for the occult nonce. For our own reflexive materialism, it should not need saying but shockingly does, is far from the only possible Western cosmology, a *final solution*. We are but medievals in waiting. And whenever panpsychism rather has periodically boomed in the Greater West—as it did definitively in the early modern era, and is beginning to do again, likewise in the face of apocalypse—so has the scientific-religious study of Magic.

The study of Islamic Occult Science is therefore essential to any postcolonialist and postcapitalist *liberation philology*.⁸³ But that is not sufficient to save the humanities, much less Western Civ, from the throes of humanity’s current existential and climatic crisis. To recover alternate Western early modernities—and hence build, perhaps, more sustainable Western postmodernities—will require a progressivist return to early modern Western deconstructive-reconstructive *cosmic philology*: the empirical history, and possibly scientific future, of *consciousness*, occult or otherwise, human or otherwise.

Notes

1. I.e., astrology, alchemy and various forms of magic and divination. Its nineteenth-century European flavor notwithstanding, I here use “occultism” to denote simply a scholarly preoccupation with one or more of the occult sciences, usually formally designated as such in Arabo-Persian classifications of the sciences, biographical dictionaries, chronicles, theological and legal tracts, etc.
2. E.g., John D. Wheeler, “Information, Physics, Quantum: The Search for Links,” in *Complexity, Entropy, and the Physics of Information*, ed. W. H. Zurek (Redwood City, Cal.: Addison-Wesley, 1990), 3–28; Vedral Vlatko, *Decoding Reality: The Universe as Quantum Information* (Oxford: Oxford University Press, 2010); David Ray Griffin, *The Reenchantment of Science: Postmodern Proposals* (Albany: SUNY Press, 1988); Thomas Nagel, *Mind and Cosmos: Why the Materialist Neo-Darwinian View of Nature Is Almost Certainly False* (Oxford: Oxford University Press, 2012); Philip Goff, *Consciousness and Fundamental Reality* (Oxford: Oxford University Press, 2017); idem, *Galileo’s Error: Foundations for a New Science of Consciousness* (New York: Pantheon Books, 2019); Elizabeth Grosz, *The Incorporeal: Ontology, Ethics, and the Limits of Materialism* (New York: Columbia University Press, 2017); John F. Haught, *The New Cosmic Story: Inside Our Awakening Universe* (New Haven: Yale University Press, 2017); Jeffrey J. Kripal, *The Flip: Epiphanies of Mind and the Future of Knowledge* (New York: Bellevue Literary Press, 2019). For a review see Joanna Leidenhag, “The Revival of Panpsychism and Its Relevance for the Science-Religion Dialogue,” *Theology and Science* 17, no. 1 (2019): 90–106.
3. It has also proved perennially popular among (Far) Western thinkers more generally even over the last few centuries. Among a host of these Jason Ā. Josephson-Storm lists the following (*The Myth of Disenchantment: Magic, Modernity, and the Birth of the Human Sciences* [Chicago: University of Chicago Press, 2017], 305): Spinoza, Leibniz, Goethe, Schopenhauer, Margaret Cavendish, Julien La Mettrie, Gustav Fechner, Ernst Mach, Henry David Thoreau, C. S. Peirce, William James, Josiah Royce, John Dewey, Henri Bergson, Samuel Alexander, Charles Strong, Pierre Teilhard de Chardin, Alfred North Whitehead, Charles Hartshorne, Albert Schweitzer, Arthur Koestler and Gregory Bateson. Elizabeth Grosz, for her part, prefers to panpsychism the terms *extramaterialism* and *incorporeality*, and calls for “a new new materialism” and “a paradoxical dualist monism” (*The Incorporeal*, 5, 13–14, 249).
4. Matthew Melvin-Koushki, “*Tahqīq* vs. *Taqlīd* in the Renaissances of Western Early Modernity,” *Philological Encounters* 3, nos. 1–2 (2018): 193–249.
5. Stephen Jay Gould, *Rocks of Ages: Science and Religion in the Fullness of Life* (New York: Ballantine, 1999).
6. H. Allen Orr, “Gould on God: Can Religion and Science Be Happily Reconciled?” *Boston Review*, October/November (1999): 33–8.
7. See e.g. Justin Stearns, *Revealed Science: The Natural Sciences in Islam in the Age of al-Ḥasan al-Yūsī (d. 1691)*, forthcoming.
8. Matthew Melvin-Koushki, “Powers of One: The Mathematicalization of the Occult Sciences in the High Persianate Tradition,” *Intellectual History of the Islamicate World* 5, no. 1 (2017): 127–99.
9. E.g., by the tenth-century Brethren of Purity; Ikhwān al-Ṣafāʾ, *Epistles of the Brethren of Purity: On Magic I*, ed. and trans. Godefroid de Callataÿ and Bruno Halflants (Oxford: Oxford University Press, 2011), 95.
10. This term is now coming into use among parapsychologists; see e.g. Dean Radin, *Real Magic: Ancient Wisdom, Modern Science, and a Guide to the Secret Power of the Universe* (New York: Harmony, 2018), and below.
11. Matthew Melvin-Koushki and James Pickett, “Mobilizing Magic: Occultism in Central Asia and the Continuity of High Persianate Culture under Russian Rule,” *Studia Islamica* 111, no. 2 (2016): 231–84, esp. 263–64.

12. Bruno Latour, *We Have Never Been Modern*, trans. Catherine Porter (Cambridge, Mass.: Harvard Univ. Press, 1993); Wouter J. Hanegraaff, *Esotericism and the Academy: Rejected Knowledge in Western Culture* (Cambridge: Cambridge University Press, 2012).
13. Hanegraaff, *Esotericism*, ch. 3; Randall Styers, *Making Magic: Religion, Magic, and Science in the Modern World* (Oxford: Oxford University Press, 2004).
14. While the Yates thesis, in its extreme form, is now generally rejected, her primary insight still holds, and has proved especially productive for history of science and philosophy. See e.g. Kevin Killeen and Peter J. Forshaw, eds., *The Word and the World: Biblical Exegesis and Early Modern Science* (Houndsmill, Basingstoke: Palgrave Macmillan, 2007); Wouter J. Hanegraaff, "Beyond the Yates Paradigm: The Study of Western Esotericism between Counterculture and New Complexity," *Aries* 1, no. 1 (2001): 5–37.
15. See e.g. Michael Taussig, *The Magic of the State* (New York: Routledge, 1997); Christopher Bracken, *Magical Criticism: The Recourse of Savage Philosophy* (Chicago: University of Chicago Press, 2007); Jeffrey J. Kripal, *Authors of the Impossible: The Paranormal and the Sacred* (Chicago: University of Chicago Press, 2010); Josephson-Storm, *The Myth*.
16. Wouter J. Hanegraaff, "Textbooks and Introductions to Western Esotericism," *Religion* 43, no. 2 (2003): 178–200.
17. Sonja Brentjes, "Between Doubts and Certainties: On the Place of History of Science in Islamic Societies within the Field of History of Science," *NTM: Zeitschrift für Geschichte der Wissenschaften, Technik und Medizin* 11, no. 2 (2003): 65–79.
18. See e.g. Edward F. Kelly, Adam Crabtree and Paul Marshall, eds., *Beyond Physicalism: Toward Reconciliation of Science and Spirituality* (Lanham: Rowman & Littlefield, 2015); Radin, *Real Magic*; cf. Egil Asprem, "Dis/unity of Knowledge: Models for the Study of Modern Esotericism and Science," *Numen* 62 (2015): 538–67. Historians of magic, however, have yet to incorporate such findings into their research. As Edward Bever notes (*The Realities of Witchcraft and Popular Magic in Early Modern Europe: Culture, Cognition, and Everyday Life* [New York: Palgrave Macmillan, 2008], 439): "It is understandable, but at the same time lamentable, that in the past 50 years of modern social-scientific investigations of early modern witchcraft and magic not one has taken into account the body of serious scientific work on paranormal phenomena that has been done in the past century, along with the vigorous skeptical critique of it that has been ongoing."
19. Melvin-Koushki, "Powers of One." To counteract entrenched colonialist-orientalist narratives as to the essential un-Westernness of Islam, I propose that the broadest possible cultural-geographical definition of the West be consistently used when writing comparative early modern intellectual history: the half of Afro-Eurasia west of South India, incorporating the Arabic, Persian and Latin cosmopolises, that vast realm where the Hellenic-Abrahamic synthesis reigned supreme, and philosophy was pursued in simultaneously mathematical-linguistic terms (Melvin-Koushki, "*Tahqīq* vs. *Taqlīd*"). To the extent that parts of Southeast Asia were fully incorporated into Islamdom during this period, moreover, by this definition they are to be considered Western.
20. Sonja Brentjes, "The Prison of Categories—'Decline' and Its Company," in *Islamic Philosophy, Science, Culture, and Religion: Studies in Honor of Dimitri Gutas*, ed. Felicitas Opwis and David Reisman (Leiden: Brill, 2012), 131–56.
21. A classic example is Armand Abel, "La place des sciences occultes dans la décadence," In *Classicism et déclin culturel dans l'histoire de l'Islam*, ed. Robert Brunschvig and Gustave von Grunebaum (Paris: Éditions Besson Chantemerle, 1957), 291–311.
22. Dallal, *Islam*, 144–45.
23. The fault for this lies not with Dallal, of course, who is simply summarizing the state of the field, relying on George Saliba's findings in particular (see e.g. his *Islamic Science and the Making of the European Renaissance* [Cambridge, Mass.: MIT Press, 2007]); and like Saliba, his intent is precisely to defend the honor of "postclassical" Islamicate science against the depredations of whiggish triumphalism. As for Ghazālī and the supposed decline of Islamic philosophy, long touted by orientalist scholars: several decades of research have now shown that, far from crippling philosophy, Ghazālī's own approach was profoundly

philosophical, and contributed not a little to that discipline's flourishing in the post-Mongol era.

24. Melvin-Koushki, "Powers of One"; idem, "In Defense of Geomancy: Šaraf al-Dīn Yazdī Rebuts Ibn Ḥaldūn's Critique of the Occult Sciences," in *Islamicate Occultism: New Perspectives*, ed. Matthew Melvin-Koushki and Noah Gardiner, special double issue of *Arabica*, 64, nos. 3–4 (2017): 346–403.
25. Dallal, *Islam*, 172.
26. Ullmann, *Die Natur- und Geheimwissenschaften im Islam* (Leiden: Brill, 1972), 1.
27. Sezgin, *Geschichte des arabischen Schrifttums*, Bd. 7: *Astologie, Meteorology und Verwandtes bis ca. 430 H.* (Leiden: Brill, 1979), 5 n. 2.
28. Richard Lemay, "L'Islam historique et les sciences occultes," *Bulletin d'Études Orientales* 44 (1993): 19–37, esp. 20. Kennedy, for instance, habitually refers to astrology as a pseudoscience, worth studying only because it involves non-trivial mathematical skill, and wholly ignores closely related sciences like geomancy; see e.g. "The Astrological Houses as Defined by Medieval Islamic Astronomers," repr. in E. S. Kennedy, *Astronomy and Astrology in the Medieval Islamic World*, xv (Aldershot: Ashgate, 1998), 535.
29. Sonja Brentjes and Robert G. Morrison, "The Sciences in Islamic Societies," in *The New Cambridge History of Islam*, Vol. 4: *Islamic Cultures and Societies to the End of the Eighteenth Century*, ed. Robert Irwin (Cambridge: Cambridge University Press, 2010), 564–693, esp. 564.
30. E.g., Jan P. Hogendijk and Abdelhamid I. Sabra, eds., *The Enterprise of Science in Islam: New Perspectives* (Cambridge, Mass.: MIT Press, 2003); Rivka Feldhay and F. Jamil Ragep, eds., *Before Copernicus: The Cultures and Contexts of Scientific Learning in the Fifteenth Century* (Montreal & Kingston: McGill-Queen's University Press, 2017).
31. This observation is based on a preliminary survey of *Index Islamicus* and history of science, technology and medicine bibliographies, especially those published annually by *Isis*.
32. Sebastian Günther and Dorothee Pielow, eds., *Die Geheimnisse der oberen und der unteren Welt: Magie im Islam zwischen Glaube und Wissenschaft* (Leiden: Brill, 2019). More generally, and despite an often high degree of philological rigor and a broad geographical and temporal scope, the volume approaches its topic primarily from a history and anthropology of religion standpoint; history of science is wholly excluded. For an extended review and critique see Matthew Melvin-Koushki, "Magic in Islam between Religion and Science," *Magic, Ritual, and Witchcraft* 14, no. 2 (2019): 255–87.
33. Bernd-Christian Otto proposes a discourse-historical approach as the best means of combating this pervasive methodological error in the study of Islamic Magic; see his "Magie im Islam: Eine diskursgeschichtliche Perspektive," in *Die Geheimnisse der oberen und der unteren Welt*, ed. Günther and Pielow, 515–46.
34. Hanegraaff, *Esotericism*, 295–314.
35. While often philologically sloppy, Corbin nevertheless spearheaded the edition and translation of primary sources; and Scholem was a consummate philologist. See Maria E. Subtelny, "History and Religion: The Fallacy of Metaphysical Questions (A Review Article)," *Iranian Studies* 36, no. 1 (2003): 91–101, esp. 100; Wouter J. Hanegraaff, *Esotericism and the Academy: Rejected Knowledge in Western Culture* (Cambridge: Cambridge University Press), 296–98.
36. Corbin, *En Islam iranien: Aspects spirituels et philosophiques*, 4 vols. (Paris: Gallimard, 1972), 3:233–74.
37. Melvin-Koushki, "Powers of One." Alidism, or devotion to the the Prophet Muḥammad's cousin and son-in-law 'Alī b. Abī Ṭālib, the Prophet's daughter and the latter's wife and the line of Imams produced from their union, was a common feature of Islamicate culture for Shi'i and Sunni alike, especially in the post-Mongol period, and especially associated with the occult sciences of lettrism and alchemy.
38. Ibid.; cf. Saliba, *Islamic Science*.
39. See e.g. Peter Dear, *Discipline & Experience: The Mathematical Way in the Scientific Revolution* (Chicago: University of Chicago Press, 1995).

40. To be clear, both developments were a few centuries in the making, but did not conceptually and socially converge until the early fifteenth century. A major upshot of this convergence was the construction of the Samarkand Observatory in 1420; the star tables (*zīj*) made there were the most accurate ever produced in the Western tradition to that date, and used and updated throughout the Islamicate and Christianate realms alike until the nineteenth century. See Matthew Melvin-Koushki, "The New Brethren of Purity: Ibn Turka and the Renaissance of Neopythagoreanism in the Early Modern Persian Cosmopolis," in *Companion to the Reception of Pythagoras and Pythagoreanism*, ed. Aurélien Robert, Irene Caiazzo and Constantin Macris, 2 vols. (Leiden: Brill, forthcoming 2020).
41. This is due, again, largely to the efforts of Yates, whose oeuvre in general, and *Giordano Bruno and the Hermetic Tradition* (1964) in particular, insisted on Christian kabbalah's relevance to Renaissance and hence Scientific Revolution intellection. By contrast, Scholem's thoroughlygoing—if usually implicit—religionist approach, which continues to structure the field to the present, has served to largely remove kabbalah's originary and contemporary Jewish iterations from history of science.
42. E.g. Kevin Killeen and Peter J. Forshaw, eds., *The Word and the World: Biblical Exegesis and Early Modern Science* (Houndsmill, Basingstoke: Palgrave Macmillan, 2007); Håkan Håkansson, *Seeing the Word: John Dee and Renaissance Occultism*. Lund: Lunds Universitet, 2001.
43. Cf. Sajjad H. Rizvi, "Philosophy as a Way of Life in the World of Islam: Applying Hadot to the Study of Mullā Ṣadrā Shīrāzī (d. 1635)," *Bulletin for the School of Oriental and African Studies* 75, no. 1 (2012): 33–45.
44. Hanegraaff, *Esotericism*, 277–95.
45. E.g., Peter Struck, *Divination and Human Nature: A Cognitive History of Intuition in Classical Antiquity* (Princeton: Princeton University Press, 2016).
46. Ironically, this psychologizing theory of magic seems to have been original to Plato himself—despite the fact that Neoplatonism would constitute the theoretical bedrock for its pursuit in the Western, Islamo-Judeo-Christian tradition until the modern period; see Fritz Graf, "Theories of Magic in Antiquity," in *Magic and Ritual in the Ancient World*, ed. Paul Mirecki and Marvin Meyer (Leiden: Brill, 2002), 93–104.
47. Cf. Hanegraaff's similar call for *methodological agnosticism* ("Empirical Method in the Study of Esotericism," *Method & Theory in the Study of Religion* 7, no. 2 [1995]: 99–129).
48. These include Edward Lane, Ignaz Goldziher, Duncan Black MacDonald, Alfred Wallis Budge, Bess Allen Donaldson, Hellmut Ritter, Martin Plessner, Paul Kraus, Shaul Shaked, Richard Winstedt, Toufic Fahd, Franz Rosenthal and others mentioned above.
49. Cf. Brentjes, "Between Doubts and Certainties," 67–68; Stearns, *Revealed Science*.
50. A case in point is Robert G. Morrison, *Islam and Science: The Intellectual Career of Nizām al-Dīn al-Nisābūrī* (New York: Routledge, 2007), which demonstrates precisely the epistemological and sociological inextricability of the emic and etic categories science and religion in the post-Mongol period.
51. For a review of earlier scholarly pushes in the same direction, see Edgar W. Francis, "Magic and Divination in the Medieval Islamic Middle East," *History Compass* 9, no. 8 (2011): 622–33.
52. Hence the spate of dissertations on various occult sciences produced in the last several years or forthcoming shortly, none of which have yet appeared in book form; see the following footnote for examples. The anthropologists are more advanced in this respect; see e.g. Alireza Doostdar, *The Iranian Metaphysicals: Explorations in Science, Islam, and the Uncanny* (Princeton: Princeton University Press, 2018) and Anand Vivek Taneja, *Jinnealogy: Time, Islam, and Ecological Thought in the Medieval Ruins of Delhi* (Stanford: Stanford University Press, 2018). But the upcoming symposium "Islamic Occult Studies on the Rise," organized by myself and Noah Gardiner at the University of South Carolina (rescheduled to Spring 2021 due to COVID-19), will feature the work of over a dozen new historians committed to this tiny field, which bodes well indeed for its continued expansion over the next decade.
53. Thus Nicholas G. Harris, "Better Religion through Chemistry: Aydemir al-Jildaki and Alchemy under the Mamluks" (Ph.D. dissertation, University of Pennsylvania, 2020), for

- example, is the first monograph on Mamluk alchemy, and the first on its Ottoman continuation is Tuna Artun, “Hearts of Gold and Silver: Production of Alchemical Knowledge in the Early Modern Ottoman World” (Ph.D. dissertation, Princeton University, 2013); Ahmet Tunç Şen, “Astrology in the Service of the Empire: Knowledge, Prognostication, and Politics at the Ottoman Court, 1450s–1550s” (Ph.D. dissertation, University of Chicago, 2016) is the first on the topic of astrology at the Ottoman court, where the science was institutionalized to an unparalleled extent; Emin Lelić, “Ottoman Physiognomy (*‘Ilm-i Firâset*): A Window into the Soul of an Empire” (Ph.D. dissertation, University of Chicago, 2017) the first on this staple imperial science; Noah D. Gardiner, “Esotericism in a Manuscript Culture: Aḥmad al-Būnī and His Readers through the Mamlūk Period” (Ph.D. dissertation, University of Michigan, 2014) and Jean-Charles Coulon, “La magie islamique et le ‘corpus bunianum’ au Moyen Âge” (Ph.D. dissertation, University of Paris IV – Sorbonne, 2013) the first to riddle the hugely influential but textually bewildering Būnian lettrist corpus; and Matthew Melvin-Koushki, “The Quest for a Universal Science: The Occult Philosophy of Ṣā’ib al-Dīn Turka Iṣfahānī (1369–1432) and Intellectual Millenarianism in Early Timurid Iran” (Ph.D. dissertation, Yale University, 2012) and idem, “Persianate Geomancy from Ṭūsī to the Millennium: A Preliminary Survey,” in *Occult Sciences in Premodern Islamic Culture*, ed. El-Bizri and Orthmann, (Beirut: Orient-Institut Beirut, 2018), 151–99, the first on the high lettrist and geomantic tradition respectively in the post-Mongol Persianate world.
54. Lynn Thorndike (d. 1965) was the first historian of Western (read here: non-Islamic) science to empirically treat of the occult sciences as mainstream within and integral to its development, at least through the 17th century; see Lynn Thorndike, *A History of Magic and Experimental Science*, 8 vols. (New York: Columbia Univ. Press, 1934–58). Given the very different, neoliberal structure of the academy today, however, an Islamicist Thorndike is unlikely to arise anytime soon.
 55. Melvin-Koushki, “Powers of One”; cf. Sonja Brentjes, *Teaching and Learning the Sciences in Islamicate Societies (800–1700)* (Turnhout: Brepols, 2018), ch. 7.
 56. Matthew Melvin-Koushki, “Early Modern Islamicate Empire: New Forms of Religiopolitical Legitimacy,” in *The Wiley-Blackwell History of Islam*, ed. Armando Salvatore, Roberto Tottoli and Babak Rahimi (Hoboken: Wiley-Blackwell, 2018), 353–75; Cornell Fleischer, “Ancient Wisdom and New Sciences: Prophecies at the Ottoman Court in the Fifteenth and Early Sixteenth Centuries,” in *Falnama: Book of Omens*, ed. Massoumeh Farhad and Serpil Bağcı (London: Thames & Hudson, 2009), 232–43.
 57. It should be noted that Ibn Sīnā’s *Treatise on the Divisions of the Rational Sciences* (*Risāla fi Aqsām al-‘ulūm al-‘aqliyya*), Ibn al-Akfānī’s own model, remained even more popular in Mamluk and Ottoman madrasa circles, and under the title *De divisionibus scientiarum* became equally seminal in Latinate Europe (my thanks to Sonja Brentjes for this observation).
 58. Melvin-Koushki, “Powers of One,” 172–73.
 59. Melvin-Koushki, “Powers of One,” 148–51; on this sanctification process generally see Gardiner, “Esotericism.”
 60. Melvin-Koushki, “The New Brethren”; Fleischer, “Ancient Wisdom.”
 61. Harris’s “Better Religion” is the first monograph on this major Mamluk occult scientist, far more influential than, say, the celebrated Paracelsus (d. 1541), who has yet been subject of dozens of excellent studies to date.
 62. Melvin-Koushki, “Powers of One.”
 63. Matthew Melvin-Koushki, “Of Islamic Grammatology: Ibn Turka’s Lettrist Metaphysics of Light,” *al-‘Uṣūr al-Wuṣṭā: Journal of Middle East Medievalists* 24 (2016): 42–113.
 64. Melvin-Koushki, “Powers of One”; on the similar pairing of kabbalah and alchemy by contemporary Latin scholars see Peter J. Forshaw, “*Cabala Chymica* or *Chemia Cabalistica*—Early Modern Alchemists and Cabala,” *Ambix* 60 (2013): 361–89.
 65. Melvin-Koushki and Pickett, “Mobilizing Magic.”
 66. Melvin-Koushki, “*Tahqiq* vs. *Taqlid*”; cf. Brentjes, *Teaching and Learning*, 10, 74–7.

67. Charles Burnett, "The Certitude of Astrology: The Scientific Methodology of al-Qabīṣi and Abū Ma'shar," *Early Science and Medicine* 7, no. 3 (2002): 198–213, esp. 210, 212.
68. Dimitri Gutas, "Certainty, Doubt, Error: Comments on the Epistemological Foundations of Medieval Arabic Science," *Early Science and Medicine* 7, no. 3 (2002): 276–89, esp. 279–80.
69. Dimitri Gutas, "The Empiricism of Avicenna," *Oriens* 40 (2012): 391–436; Jon McGinnis, *Avicenna* (Oxford: Oxford University Press, 2010), 47–52.
70. As 'Alī Ṣafī Kāshifī (d. 1533), son of the great Timurid polymath and occultist Ḥusayn Vā'iz Kāshifī (d. 1505), asserts in his programmatic lettrist manual *Amulet of Protection from the Trials of Time* (*Hirz al-amān min fitan al-zamān*), representing the state of the art at the turn of the 16th century (trans. in Melvin-Koushki, "The Quest," 279): "Experiential essaying aimed at manifesting the effects and properties of the letters has often been successfully repeated; and experience is among the forms of proof that produce certainty (*tajriba yaki az isbāt-i yaqīn ast*)."
71. Gardiner, "Esotericism"; Matthew Melvin-Koushki, "How to Rule the World: Occult-Scientific Manuals of the Persian Cosmopolis," *Journal of Persianate Studies* 11, no. 2 (2018): 140–54.
72. Josephson-Storm, *The Myth*, 44–51.
73. Dimitri Gutas, "Avicenna and After: The Development of Paraphilosophy. A History of Science Approach," in *Islamic Philosophy from the 12th to the 14th Century*, ed. Abdelkader Al Ghouz (Göttingen: V&R unipress/Bonn University Press, 2018), 19–71.
74. Alireza Doostdar, *The Iranian Metaphysicals: Explorations in Science, Islam, and the Uncanny* (Princeton: Princeton University Press, 2018).
75. Melvin-Koushki, "Islamicate Empire." Cf. Josephson-Storm's observation that Bacon's famous phrase has been routinely misunderstood; in context, it means rather "[God's] knowledge is itself power" (*The Myth*, 46–47). As that may be, it must not be forgotten that the primary objective of our early modern Muslim occult philosophers was precisely *theosis*, aka *theomimesis*, which drove in turn their projects to help imperially realize, at least as a temporary manifestation, the Neopythagorean One in time and space.
76. Otto, "Magie im Islam," 533–6.
77. Kripal, *The Flip*, 36–7.
78. The Large Hadron Collider at CERN is here exhibit A (*ibid.*, 38).
79. As Monica Green notes (ed., *Pandemic Disease in the Medieval World: Rethinking the Black Death* [Kalamazoo: Arc Medieval Press, 2015], 9), "the mortality caused by the Black Death is the highest of any large-scale catastrophe known to humankind, save for the impact of smallpox and measles on indigenous peoples in first-contact events of the early modern period." For plague's transformative impact on the Ottoman Empire in particular, see Nükhet Varlık, *Plague and Empire in the Early Modern Mediterranean World: The Ottoman Experience, 1347–1600* (Cambridge: Cambridge University Press, 2015).
80. Melvin-Koushki, "*Tahqīq* vs. *Taqlīd*."
81. Thomas S. Kuhn, *The Structure of Scientific Revolutions*, 4th ed. (Chicago: University of Chicago Press, 2012), 171. My call here echoes that of Stearns in *Revealed Science*.
82. Steven Shapin, *The Scientific Revolution* (Chicago: University of Chicago Press, 1996).
83. Sheldon Pollock, "Philology and Freedom," *Philological Encounters* 1 (2016): 4–30.

Acknowledgements

My thanks to Sonja Brentjes, Lissa Roberts, Nicholas Banner, Nicholas Harris, Andrew Berns and Kaveh Hemmat for their feedback on a draft of this manifesto, as well as to the audience at my similarly titled talk at Columbia University's Middle East Institute on January 30, 2019. Needless to say, all views here expressed are my own.

Disclosure Statement

No potential conflict of interest was reported by the author(s).

Note on contributor

Matthew Melvin-Koushki (PhD Yale) is Assistant Professor and McCausland Fellow of History at the University of South Carolina. He specializes in early modern Islamicate intellectual and imperial history, with a focus on the theory and practice of the occult sciences in Timurid-Safavid Iran and the broader Persianate world to the nineteenth century.